

SPARE PART CATALOG

16V4000M33S/527111561



Power. Passion. Partnership.

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

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Preface

This catalog is the basic reference document for ordering spare parts. Illustrations are provided for extensive assembly operations to assist trained personal in conducting such operations but do not claim to be complete. For better overview, even product components that are not available as spare parts are shown.

The individual components were taken from the assembly list.

Catalog structure

The catalog is divided into Function Groups (FG) and Function Group Segments (FGS) in ascending order.

The side contain the following information:

- **MODEL:** (Product Type)
- **SERIAL NO.:**
- **FG/FGS DESIGNATION**
- **FG/FGS NO.:**
- **BOM No.:**

Explanation of columns

- **ITEM:** Provides a cross-reference between illustration and text section.
- **PART NO.:** This is the valid number for ordering the part in question.
- **DESIGNATION:** The precise part name.
- **MODEL/TYPE** is an additional specification.
- **QTY.: (Quantity in Unit):** Indicates the quantity of this part required for the case in question. Quantities not stated are available on request.
- **UNIT:** The 3-character codes indicate: PC = Piece, KG= Kilogram, M= Meter, ROL= Roll, PAC= Pack
- **WEIGHT**
- **FN.:** (Footnote) Provides additional parts information. For clarification of footnote symbols, see Page 2.
- **DIMENSIONS:** This column contains standard-part dimensions or manufacturer designations for purchased parts.
- **STANDARD**

Amendments service

Part No. changes and amendments which occur after publication of the catalog are not published. We therefore recommend that users make a record of amendments in their catalogs. This information is provided, for instance, in order confirmations and/or on spare parts labels.

Footnotes: Key

AB	Only available complete. Repair by manufacturer only.
AC	Not scheduled as a spare part. Availability must be checked.
ACN	Not scheduled as a spare part. Availability must be checked. Component only available complete.
EM	Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.
EMN	Emission-relevant parts. Component only available complete.
KD	Group balancing required
KE	Individual part; not scheduled as a spare part.
KF	Adjusted spare part.
KFN	Adjusted spare part. Component only available complete.
M	Spares-related components or parts with different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.
MN	Spares-related components or parts with different repair sizes. Components only available complete.
MUB	Spares-related components or Parts with different repair sizes. Available only as complete set.
NN	Component is not broken down into further sub-components.
P	List of bought-in components
SB	Group of parts which are welded or brazed together
UA	Indicate dimensions when ordering
UB	Only supplied complete or as a complete set.

Ordering Procedure

The following information is essential for spare-part order processing:

- Engine Model
- Serial number (see engine name plate)
- FG/FGS
- Item
- Part No.
- Designation
- Model/Type
- Quantity
- Page

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

When ordering a calibrated part (marked by "KF", "KFN" in the "FN" column), type the following additional information in the **Comments** field:

If the calibration No. is known (see part's nameplate):

Add the calibration No., preceded by an oblique stroke, as an extension to the part number ("/xxxxx")

e.g.: 0025362810/E1234 Here, the extension "E1234" would be the part's calibration number on the nameplate.

If the calibration No. is not known:

Give a detailed description of the required part's installation location by quoting the complete Device Description code (e.g. "= 37.1.1 + A022 - F003") plus the part number extension "/77" (e.g. 5005362810/77).

In this case, the calibrated spare part will be supplied under a new part number.

The Device Description is found on the nameplate at the part's installation location, or in the corresponding Wiring Diagram.

When ordering a non-calibrated adjustable part (marked by "KF", "KFN" in the "FN" column) which should not be calibrated by the factory: Add extension "/76" to the part number (e.g. 0025362810/76).

Phone orders should be confirmed in writing to prevent errors.

Ordering Example:

Engine model	Serial number	FG/FGS	ITEM	Part No.	Designation	Quantity	Page
16V4000R41	527101836	031 88	5	5270300701	Crankshaft	1	33
		083 05	10	0020922801	Easy-Cange Filter	2	92
Blue Line	V4000M Waterjet	517 00	75	5185301593	Control Console	1	53

Factory repairs

Parts shipped to MTU for repair must be marked with the relevant engine serial number (tag or label, for example).

Exchange components (Reman)

Exchange components are parts overhauled by the manufacturer. They have the same service life and quality characteristics as new parts.

Components which are included in the Reman exchange program are marked by the following letters in front of the item number:

For instance: **E 524 010 3520 = New component in exchange program**

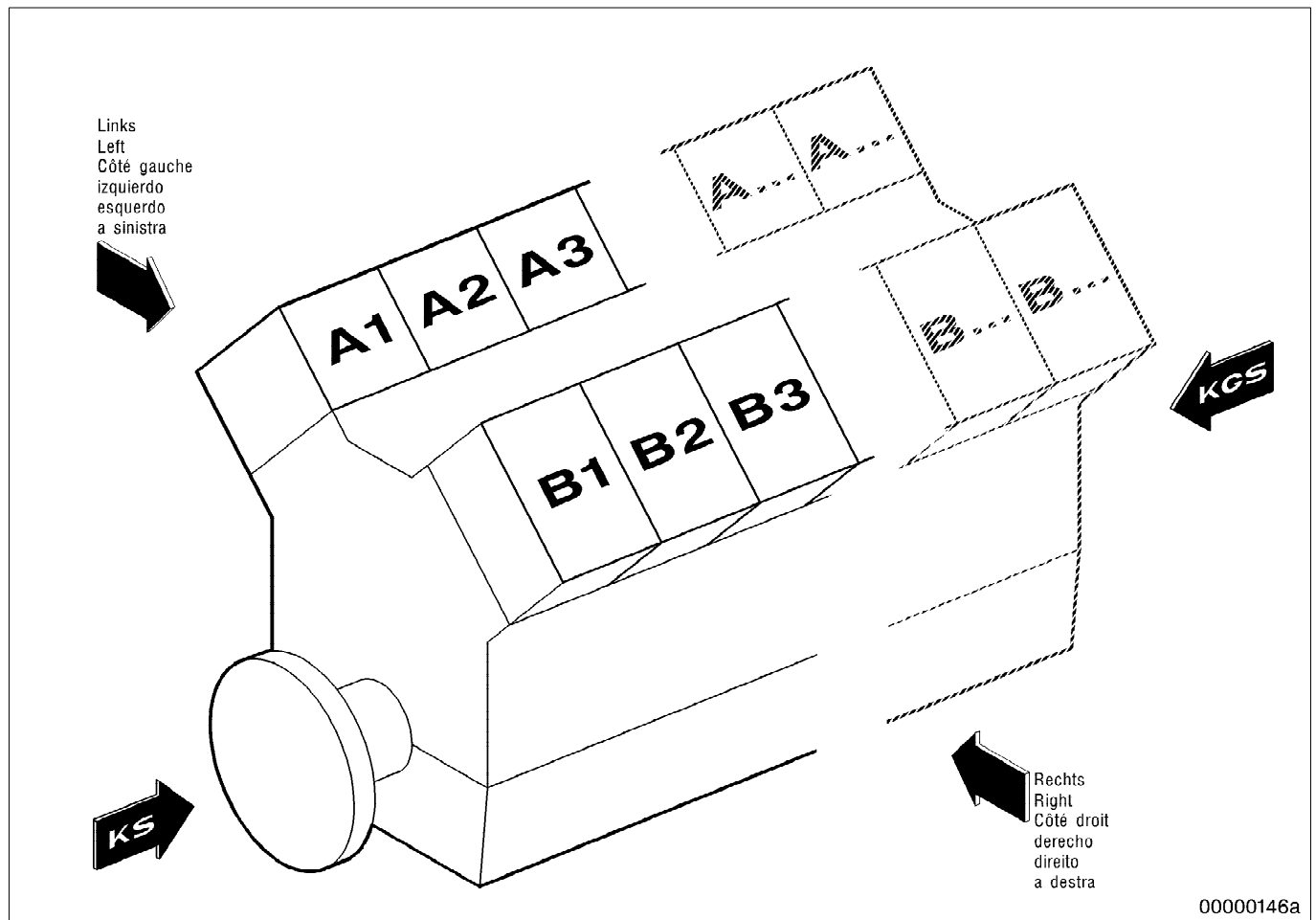
R 524 010 3520 = Reman component in exchange program

Exchange procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

Further information upon request.

Engine side and cylinder designations



Engine sides / banks are always designated as viewed from the driving end (KS).

The left bank of cylinders is designated "A" and the right bank "B" (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively from the driving end, starting with No. 1.

Other engine components are also numbered consecutively from the driving end, starting with No. 1.

Designations and abbreviations

Driving end = KS

Free end = KGS

Left-hand side

Right-hand side

Top

Bottom

Turbocharger = ATL

FG FG DESIGNATION

- 004 LABELS AND AUXILIARY MATERIALS
- 010 CRANKCASE AND ATTACHMENTS
- 011 CRANKCASE
- 014 OIL PAN
- 015 ENGINE MOUNTING
- 016 END HOUSING, DRIVING END AND FREE END
- 018 CRANKCASE BREATHER
- 024 GEARCASE, AUX PTO END
- 030 RUNNING GEAR
- 031 CRANKSHAFT WITH BEARINGS
- 032 CRANKSHAFT, PTO ON DRIVING END
- 035 CRANKSHAFT, PTO ON AUX PTO END
- 036 TRANSPORT LOCK/LOCK FOR CRANKSHAFT
- 037 PISTON
- 038 CONNECTING ROD
- 041 CYLINDER HEAD
- 051 CAMSHAFT
- 055 VALVE DRIVE
- 073 INJECTION PUMP / HP PUMP
- 075 FUEL INJECTOR
- 077 INJECTION LINE / HP FUEL LINE
- 081 FUEL PIPEWORK BEFORE DELIVERY PUMP
- 082 FUEL PIPEWORK FROM DELIVERY PUMP TO FILTER
- 083 FUEL FILTER
- 086 LEAK-OFF FUEL LINE / FUEL RETURN
- 087 LEAK-OFF FUEL MONITORING
- 101 EXHAUST TURBOCHARGER, GENERAL OR H.P. TURBOCHARGER
- 111 CHARGE AIR COOLING, GENERAL, LEFT SIDE
- 120 AIR SYSTEM / AIR DUCTING
- 121 AIR FILTER
- 122 AIR INTAKE SILENCER
- 124 AIR SUPPLY TO CYLINDERS
- 125 AIR ROUTING BETWEEN TURBOCHARGER AND CHARGE-AIR COOLER
- 140 EXHAUST SYSTEM
- 141 EXHAUST SYSTEM AFTER CYLINDER HEAD
- 143 CARRIER HOUSING
- 145 EXHAUST SYSTEM AFTER TURBOCHARGER
- 146 EXHAUST DUCTING AFTER ENGINE (TO END OF EXH. AFTERTREATMENT)
- 172 ELECTRIC STARTING
- 173 AIR STARTER
- 180 LUBRICATING OIL SYSTEM
- 181 LUBE OIL PUMP AND DRIVE
- 183 OIL FILTRATION/COOLING
- 184 CRANKCASE-INTERNAL OIL SUPPLY
- 185 TURBOCHARGER OIL SUPPLY
- 188 CONNECTING HARDWARE FOR ENGINE OIL SUPPLY

FG FG DESIGNATION

- 189 OIL LEVEL REGULATING PUMP
- 200 COOLANT SYSTEM
- 202 COOLANT CIRCUIT, GENERAL, HT CIRCUIT
- 203 LOW-TEMPERATURE CIRCUIT
- 205 COOLANT COOLER ATTACHING H/WARE
- 206 COOLANT PIPEWORK WITH THERMOSTAT
- 207 RAW WATER PUMP WITH CONNECTIONS
- 208 COOLANT EXPANSION TANK
- 231 ENGINE MOUNTING
- 252 RUBBER COUPLING
- 293 OIL PUMP
- 301 GENERAL ACCESSORIES
- 500 MONITORING AND CONTROL SYSTEM, GENERAL ELECTRICAL EQUIPMENT
- 507 WIRING, GENERAL, FOR ENGINE/GEARBOX/PLANT
- 508 ACCESSORIES FOR ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM
- 515 PRESSURE/TEMPERATURE MONITORING SYSTEMS (EXCL. EXHAUST GAS)